Unit 2

Target Vocabulary Unit 2.1 Recent breakthroughs

1. **beyond** *(prep)* - past a certain point, limit or stage. *Space telescopes allow us to see beyond our own galaxy. NEO became the world's first invasive BCI approved for use beyond clinical trials.*
2. **breakthrough** /ˈbreɪkθruː/ (n) - an important discovery or event that helps to improve the situation or provide an answer to a problem. *The NEO brain-computer interface is considered a major breakthrough in medicine.*
   breakthrough (adj) – very important in the progress or development of something or someone, because of solving a big problem or making a big improvement. *A new breakthrough technology called memristor-based computing-in-memory (CIM) eliminates the need for energy-intensive data transfer.*
3. **conduct** /kənˈdʌkt/ (v) – to organize and perform a particular activity. *With lab-grown human skin, experiments can be conducted without animal testing. The first clinical use of the personalized cancer vaccine was conducted in April 2026 on a 60-year-old patient with skin melanoma in Russia.*
4. **credit smb with V-ing** /ˈkredɪt/ (focus on the people/agent) – to say that someone is responsible for something good. *The researchers from the University of Portsmouth are credited with discovering an enzyme that can digest plastic. Australian scientists from the University of Queensland are credited with growing human skin in a laboratory.*
   credit smt to smb (focus on the achievement) *Lab-grown human skin is credited to Australian scientists from the University of Queensland.*
5. **discover** /dɪˈskʌvər/ (v) – to find information, a place, or an object, especially for the first time. *Alexander Fleming discovered penicillin by accident.*
   discovery /dɪˈskʌvəri/ (n) – the process of finding information, a place, or an object, especially for the first time. *The discovery of oil brought many benefits to the town.*
   accidental/ big/ great/ important/ major/ remarkable/ scientific/ significant discovery
6. **eliminate** (v) - to remove or take away something. *Vaccines help eliminate dangerous diseases around the world. The NEO brain-computer interface eliminates the risks of bleeding and scarring associated with traditional brain implants.*
7. **innovate** (v) – to introduce changes and new ideas to the existing product. *Scientists continue to innovate in the field of cancer treatment, developing vaccines like EnteroMix.*
   innovation (n) – a change or improvement made to an existing product, idea, or method. *The smartphone was one of the most significant innovations of the 21st century.*
8. **invent** (v) - to design or create something that has never been made before. *The World Wide Web was invented in 1989.*
   invention (n) – something that has never been made before. *The world changed rapidly after the invention of the phone.*
9. **lack** *(v)* - to not have something. *Past artificial skin (dating back to the 1970s) lacked blood vessels. Most eVTOLs lack the ability to fly long distances.*
10. **reduce** (v) - to make something less in size, amount, importance, price, etc. *There is hope that hydrogen-powered aircraft will be built to reduce air pollution.*
11. **vital** /ˈvaɪtəl/ (adj) – necessary for the success, extremely important. *eVTOLs are especially vital for crowded cities facing air pollution.*

Target Vocabulary Unit 2.2 Innovation and technology

1. **bacterium** /bækˈtɪəriəm/ **bacteria** (plural) /bækˈtɪəriə/ (n) – the simplest and smallest forms of life. *Bacteria are small single-celled organisms.*
2. **deplete** /dɪˈpliːt/ (v) – to reduce something in size or amount. *If we continue to deplete the earth's natural resources, we will cause serious damage to the environment.*
   deplete heavily/ seriously/ severely
   syn. exhaust
3. **essential** /ɪˈsenʃəl/ (adj) – extremely important. *Protecting eyes from dangerous radiation is essential.*
   essential for/ to
4. **harmful /ˈhɑːmfəl/ (adj) – causing damage or injury. *This group of chemicals is known to be harmful.***
   harmful to
   extremely/ very/ potentially/ environmentally harmful
5. **influence** /ˈɪnfluəns/ (v) – to have an effect on something/ someone. *The discovery of penicillin has influenced modern medicine.*
   influence somebody/ something
   considerably/ greatly/ strongly influence
   influence (n) **–** the effect that somebody/ something has on somebody/ something else. *The influence of biomimicry can be seen in many everyday products.*
   to have an influence on/ be under an influence
   big/ enormous/ great/ major/ negative influence
6. **initially** /ɪˈnɪʃəli/ (adv) – at the beginning. *His invention was initially unpopular with users.*
   initial (adj) – the first, happening at the beginning
7. **injury** /ˈɪndʒəri/ (n) – physical harm or damage. *The device protects your insides from injury.*
   bad/fatal/ visible/ serious/ terrible injury
   cause/ suffer from an injury
   injure /ˈɪndʒər/ (v) – to hurt or cause physical harm. *She fell and injured her shoulder.*
   injure accidentally/ badly/ seriously
8. **launch** /lɔːntʃ/ (v) – to make a product or service available to the public for the first time. *The website was launched in July.*
   launch *(n)* – the introduction of a product or service. *A rocket launch is scheduled for September, 11.*
   official/ public/ successful/ product launch
   get something ready for/ prepare something for launch
   launch date/party
9. **manufacture** /ˌmænjəˈfæktʃər/ (v) – to produce goods in large numbers. *He works for a company that manufactures car parts.*
10. **purpose** /ˈpɜːpəs/ (n) – aim or function of something. *Inventors create products which will serve a specific purpose to help us.*
    main/ primary/ practical/ useful purpose
    achieve/ have/ lack/ serve a purpose
11. **resistance** /rɪˈzɪstəns/ (n) – a force that stops something moving or makes it move more slowly.
    air/ wind resistance
12. **resource** /rɪˈzɔːs/ (n) – a useful or valuable possession or quality of something or somebody. *They promote the use of renewable recourses like wind power.*

renewable/ sustainable/ natural/ material resource

13. **specific** /spəˈsɪfɪk/ (adj) – connected with one particular thing only. *What's the specific purpose of this device?*
    be specific
    specific about
14. **stick** /stɪk/ (v) – to fix something to something else; to become fixed to something. *We used glue to stick the broken pieces together. Burdock seeds stuck **to** the dog's hair.*
15. **surface** /ˈsɜːfɪs/ (n) – top layer of something. *Scientists have detected several dark sports on the surface of the Sun.*
    flat/ hard/ rough/ solid surface
    on a/ the surface (of)

Target Vocabulary Unit 2.3 Creative problem solving

1. **adopt** /əˈdɒpt/ (v) – to accept or start to use something new
   We need to adopt a new strategy to solve the problem.
   adopt an approach / policy / attitude etc.
2. **charge** /tʃɑːdʒ/ (v) – to put electricity into an electrical device. *It's not working, I don't think the battery is charging.*
   recharge /ˌriːˈtʃɑːdʒ/ (v) – to fill with electricity again. *When the computer beeps, you need to recharge the battery.*
3. **come up with** (phr v) – to think of an idea. *We have come up with an amazing design for our product.*
   come up with a plan / solution / idea / proposal etc.
4. **consume** /kənˈsjuːm/ (v) – to use something. *This device does not consume any electricity.*
   consume food / energy / fuel / natural resources etc.
   consumer (n) – someone who buys or uses something
5. **complex** /ˈkɒmpleks/ (adj) – made of different parts that are connected. *We live on impossibly complex planet.*
6. **entrepreneur** /ˌɒntrəprəˈnɜːr/ (n) – someone who starts their own business. *He has become a successful entrepreneur, investing in sports, entertainment and media.*
7. **generate** /ˈdʒenəreɪt/ (v) – to create. *We need someone to generate new ideas.*
   generate jobs / income / interest / electricity etc.
8. **initial** /ɪˈnɪʃəl/ (adj) – first or happening at the beginning – *We are in the initial stages of our plan.*
   initially – at the beginning. *Initially, our plan worked well.*
9. **instead of** /ɪnˈsted/ (prep) – in place of someone or something. *You can go instead of me if you want.*
10. **invest <u>in</u>** /ɪnˈvest/ (v) – to give money to a business to get profit. *The city plans to invest **<u>in</u>** its educational system this year.*
    investment /ɪnˈvestmənt/ (n) – the money that you put in a bank, business, etc in order to make a profit, or the act of doing this. *Businesses need to increase their investment in new technology.*
11. **large scale** (phr) – large in size *Manufacturing is done **<u>on</u>** a large scale.*
12. **be opposed <u>to</u>** /əˈpəʊzd/ (phr) – to disagree with a plan or activity. *She is opposed to the idea of moving abroad.*
13. **reduce** /rɪˈdʒuːs/ (v) – to make something become smaller. *Our solution will help to reduce the cost of the product.*
    reduction /rɪˈdʌkʃən/ (n) **–** the act of reducing something. *There is no reason why the new technology should mean a massive reduction in employment.*
14. **run out** (phr v) – to use all of something and not have any left. *We ran out of petrol on our way home.*
    run out <u>of</u> money / food / patience
15. **treat** /triːt/ (v) – to cure a person of a disease *Antibiotics are used to treat bacterial infections.*

treatment of / for /ˈtriːtmənt/ (n) – something that is done to cure someone who is injured or ill *There have been great advances in the treatment of cancer.*

Target Vocabulary Unit 2.4 Creativity

1. **acquire** /əˈkwaɪə/ (v) – to get a new skill or knowledge by learning it. *She has acquired a good knowledge of English.*
   acquisition /ˌækwɪˈzɪʃn/ (n) – the act of getting something, especially knowledge or a skill
2. **advance** /ədˈvɑːns/ (v) – if something advances, it develops and improves. *Technology is advancing very quickly.*
   advance /ədˈvɑːns/ (n) – progress or development in a particular area.
3. **convince smb of smt** /kənˈvɪns/ (v) – to make somebody believe that something is true. *He couldn't convince him of the seriousness of the situation.*
   convinced /kənˈvɪnst/ (adj) – completely sure about something. *I am not entirely convinced by this argument.*
   convincing /kənˈvɪnsɪŋ/ (adj) – making someone believe that something is true.
   convincing argument/explanation
4. **create** /kriˈeɪt/ (v) – to make something happen or exist. *The government plans to create more jobs for young people.*
   creation /kriˈeɪʃn/ (n) – the act or process of making something new.
   creative /kriˈeɪtɪv/ (adj) – involving the use of skill or imagination to produce something new, for example a work of art.
   creativity /ˌkriːeɪˈtɪvəti/ (n) – the use of skill or imagination to produce something new.
   creator /kriˈeɪtə(r)/ (n) – a person who has made or invented a particular thing.
   creature /ˈkriːtʃə(r)/ (n) – a living thing that can move around, such as an animal.
   living creature/ sea creature
5. **destroy** /dɪˈstrɔɪ/ (v) – to damage something so greatly that it no longer exists
   destruction /dɪˈstrʌkʃn/ (n) – damage that is so serious that something stops existing and can never return to its previous state. *The destruction of the forests leads to serious environmental problems.*
   destructive /dɪˈstrʌktɪv/ (adj) – causing very serious damage
6. **emerge** /ɪˈmɜːdʒ/ (v) – to start to exist or to become known, **syn**. – to appear. *Several facts started to emerge during the investigation.*
   emergence /ɪˈmɜːdʒəns/ (n) – the moment or the fact of emerging.
   emergence of the theory/hypothesis/the Internet
7. **evolution** /ˌiːvəˈluːʃn/ (n) – 1) the slow development of plants and animals as they adapt to changes in the environment 2) the way in which something gradually changes and develops
   cultural/biological evolution/human evolution
   evolve /ɪˈvɒlv/ (v) – to develop and change gradually, especially from a simple to a more complicated form. *Slowly, these tiny organisms evolved into very complex creatures.*
   evolutionary /iːvəˈluːʃənri/ (adj) – connected with slow development or change
8. **implement** /ˈɪmplɪment/ (v) – to make something such as an idea, plan or system start to work and be used. *The agreement was signed, but the decisions were never implemented.*
   implementation /ˌɪmplɪmenˈteɪʃn/ (n) – the process of implementing something
   the implementation of the new system/decisions/reforms
9. **innovate** /ˈɪnəveɪt/ (v) – to introduce new ideas, things or ways of doing something
   innovation /ˌɪnəˈveɪʃn/(n) – the introduction of new ideas, things or ways of doing something. *The company is very interested in product design and innovation.*
   innovator /ˈɪnəveɪtə/ (n) – a person or company that introduces new ideas, things or ways of doing something

innovative /ˈɪnəveɪtɪv/ (adj) **–** introducing or using new ideas or ways of doing something. *There will be a prize for the most innovative design.*

10. **matter** /ˈmætə(r)/ (v) **–** to be important or have an important effect on somebody or something. *The results of the experiment mattered a lot to us.*
11. **relationship** /rɪˈleɪʃnʃɪp/ (n) **–** the way in which two or more people or things connected. *Scientists study the relationship between these two processes.*
12. **transform** /trænsˈfɔːm/ (v) **–** to make someone or something completely different, to change from one form to another. *The photochemical reaction transforms the light into electrical impulses.*
    transformation /ˌtrænsfəˈmeɪʃn/ (n) **–** a complete change in somebody or something
    transformative /trænsˈfɔːmətɪv/ (adj) **–** using or able to cause an important change in somebody or something, especially the one that makes them better

Unit 2.5 Description of a process

1. **about** (adv) – a little more or less than the given number. *The size of the structure is about two square metres.* Syn. **around**
2. **alternative** (adj) /ɒlˈtɜːnətɪv/ – something you can chose instead of something; another option. *Do you have any alternative plan if something goes wrong?*
   alternatively (adv) /ɒlˈtɜːnətɪv.li/ – is used for giving another option. *Water is used for cooling. **Alternatively,** automobile anti-freeze can be used.*
4. **cause** (v) /kɔːz/ **smt/to do smt** – to make something happen, especially something bad. *The error in calculations **caused that accident**. What **caused you to change** your mind?*
   to cause an error/increase/change
   cause (n) /kɔːz/ - the reason why something, especially bad happens. We are trying to find out the causes **<u>of</u>** this error.
   cause of the problem/the change; natural cause
5. **collect** (v) – to get and keep something together. *Information is collected and stored here.*
6. **convert** (v) /kənˈvɜːt/ – to change something into something new or different. *In this process energy of light converts **<u>to</u>** heat energy.*
   to convert data, to convert heat to electricity
7. **eventually** (adv) /ɪˈventʃuəli/ – in the end. *The process took several hours but eventually all the data was copied.* Syn. **finally**
8. **heat** (n) – warmth or the quality of being hot. *This physical process generates a lot of heat.*
   heat (v) – to make something hot or to become hot. *We need to heat the oil to 75° C.*
9. **liquid** (n) /ˈlɪkwɪd/ – a substance, for example water or milk, that can flow, has no fixed shape and it is not a solid or a gas, , *a glass of white liquid/hot liquid*
   liquid (adj) /ˈlɪkwɪd/ - in the form of liquid.
   liquid nitrogen/substance
10. **once** (conj.) – from the time when something happens; as soon as. *Once the process has been finished, the liquid is heated.*
11. **outline** (v) – to give the main ideas of something (a text, a plan etc.) without giving too many details. *The document outlines our programme.*
    to outline the causes of the process, to outline key points
    outline (n) – the main ideas or facts about something.
    outline of the events, a written outline
12. **pressure** (n) – the force or weight that is put on something. *The pressure of water turns the wheel.*
    to apply/relieve pressure, be under pressure

13. **rapidly** (adv) – quickly. *Things are changing very rapidly.*
    rapid (adj) *rapid movements, a rapid increase*
14. **subject** someone/something **to** something (phrasal v) – to make someone/something experience something. *The product is **subjected to high temperature** in order to kill some possible viruses.*
    to subject to high pressure, to be subjected to examination
15. **via** (prep) /ˈvaɪə/ – by means of or by way of. *I sent the application paper via email.*

Target Vocabulary Unit 2.6

1. **definitely** /ˈdefɪnətli/ (adv) – without any doubt; certainly. *The book is definitely useful for undergraduate courses. A. Oh, do you think so? B. Definitely.*
2. **draw** /drɔː/ (drew, drawn) (v) – to cause to move toward, to pull or to bring something in. *Try to draw this box inside the room. There is a fan here in the front which draws in air.*
3. **identify** /aɪˈdentɪfaɪ/ (v) – to recognize someone or something and be able to say who they are/what it is. *Our consultants are able to quickly identify the key issues.*
   to identify oneself/priorities/needs
4. **impact** /ˈɪmpækt/ (n) – an effect, or an influence. *The anti-smoking campaign had quite an impact **on** young people.*
   to make an impact on smt/smb; to consider/assess/feel impact of smt
   impact /ɪmˈpækt/ (v) – to have an effect or influence on someone or something. *The new engine could impact (on) the way future cars are designed.*
5. **issue** /ˈɪsjuː/ (n) – a problem that needs to be considered. This is a serious security issue.
   controversial/major/key issue
6. **item** /ˈaɪtəm/ (n) – something that is part of a list or group of things. *Repairs were carried out on several items **of** equipment.*
   the first/last item on the list
7. **major** /ˈmeɪdʒə/ (adj) – important, serious, large, or great. *Citrus fruits are a major source of vitamin C.*
   major part/role/decision/finding
8. **precisely** /prɪˈsaɪsli/ (adv) – 1) exactly; *Lessons begin at nine o'clock precisely.*
   2) used for showing that you completely agree with what someone says. *'You mean he took the money for himself?' 'Precisely.'*
9. **pure** /pjʊər/ (adj) – clean and free from harmful substances or not mixed with anything else. *The mountain air was wonderfully pure.*
   pure water/color/science
   purity /ˈpjʊrəṭi/ (n) a quality or state of being clean or not mixed with anything else.
   high/low purity
   purified /ˈpjʊərɪfaɪd/ – made pure. *purified water*
   purification /ˌpjʊərɪfɪˈkeɪʃ(ə)n/ (n) – the process of making something clean by removing dirty or harmful substances from it. *an air purification system*
10. **spin** /spɪn/ (spun) (v) - to (make something) turn round and round quickly. *Spin the wheel with your hand. Why does the Earth spin?*
    spin /spɪn/ (n) – the movement of something turning around. *We calculated the rate of spin of the planet.*
11. **time-consuming** /ˈtaɪm kənsjumɪŋ/ (adj) – something that takes a long time to do. *This method requires time-consuming data interpretation.*
    time-consuming process/task
12. **various** /ˈveəriəs/ (adj) – many different. *They are seeking financial support from various sources.*
    various methods/tools/shapes